Mitchell Knoth

CLOUD SOFTWARE ENGINEER · AGENTIC ENGINEERING & DEVELOPER EXPERIENCE

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Cloud Software Engineer with 7+ years of experience within the Developer Experience realm ranging from Enterprise Networking Backbone to Agentic Engineering at scale. Current work entails Agent Outposts or Long-lived Agents at scale, secure agentic engineering via remote and local sandboxing strategies, and procuring AI insights to increase visibility into AI usage and whether folks are leveraging AI Effectively

EXPERIENCE

John Deere | Cloud Software Engineer

December 2019–Present · Des Moines, IA

SDLC-Next — AI Software Delivery R&D · April 2026–Present

- Pioneering execution platform that centralizes agentic usage both local and remote to ensure proper guardrails and audit trails are in place in what used to be blackboxed.
- Delivered full-stack AI Insights pages linking AI usage with DORA metrics giving Engineers valuable insights into how they are leveraging AI; started the app in June of 2026 and already have **~600 unique monthly users**.
- Piloting CloudflareOS to allow model diversification, subject-matter expert agents with small subset of users to challenge how we handle Context Engineering on a large enterprise scale
- Leveraging **Andrew Karpathy's LLM Wiki** I built a team knowledge repository making research and engineering guidance searchable and traversable by agents; adopted by the immediate team and pattern shared and scaled to a handful of other teams.

PR to Production — Developer Experience

- Led event-driven segregation-of-duties monitoring across **50,000+ repositories**, integrating GitHub workflow events with AWS compliance systems; delivered before the audit deadline.
- Provided architecture and code reviews, technical mentorship, and incident post-mortems across engineering teams.

OneCloud — Cloud Infrastructure & Networking

- Led network consolidation across **~2,000 VPCs**, optimizing NAT Gateway usage to reduce annual cloud spend by **$1 million**.
- Led an enterprise DNS migration from **BIND to Amazon Route 53**, addressing **140,000+ associated security findings**.
- Migrated **10,000+ AWS ACM certificates** to automated DNS validation, enabling renewal without manual email approvals.
- Automated account and VPC deletion using **AWS Step Functions and Lambda**.
- Introduced **Dev Containers** to provide reproducible developer environments and simplify onboarding.

ADDITIONAL EXPERIENCE

John Deere | Earlier Platform Contributions

License Management & Global Gateway

- Built a secure license-data ingestion prototype preserving isolation and access controls, helping make the case for a new Data Engineering team.
- Architected the API platform's first external-facing API with **role-based authorization and audit logging** for dealer access to sales leads.

Oakline | Founder & Software Engineer

March 2026–Present

- Founded and built an insurance-agency SaaS platform for **policy sales and customer records**, now in an early pilot with one agency.
- Own product, application development, **AWS infrastructure**, and pilot support; use **Cloudflare DNS** for domain management.
- Built a multi-agent workflow with test-driven implementation and specialist-agent review; maintain GitHub Actions verification covering unit, database integration, Terraform infrastructure, and browser tests, plus application and Docker builds.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java

Cloud: AWS, Terraform, CloudFormation, Docker, ECS Fargate, Cloudflare DNS

Developer tooling: GitHub Actions, CI/CD, Dev Containers, Cloudflare Sandbox, AI Gateway

Observability & data: Datadog RUM, CloudWatch, Redis, Grafana, Splunk, AWS X-Ray

Networking & security: VPC, Transit Gateway, Route 53, OAuth, certificate automation

Favorite Agent Methodologies: Implementer-Reviewer-Final Loop, Agent Swarm, Remote Agents / Agent Outposts

CERTIFICATIONS

AWS Certified Solutions Architect – Professional · November 2026

AWS Certified DevOps Engineer – Professional · May 2026

AWS Certified Advanced Networking – Specialty · August 2025

AWS Certified SysOps Administrator – Associate · February 2025

AWS Certified Solutions Architect – Associate · March 2024

AWS Certified Cloud Practitioner · November 2021

EDUCATION

Iowa State University · B.S. Software Engineering · 2020

Minor: Cyber Security · Cum Laude